

RISK & RETURN IN INVESTING

Financial Mathematics • CA-III • Unit V and Unit VI

Name: _____

PRN / Roll No: _____

Division: _____

UNIT V: BASICS OF INVESTMENT

1. Investment Return

Gain or loss as a % of the money invested.

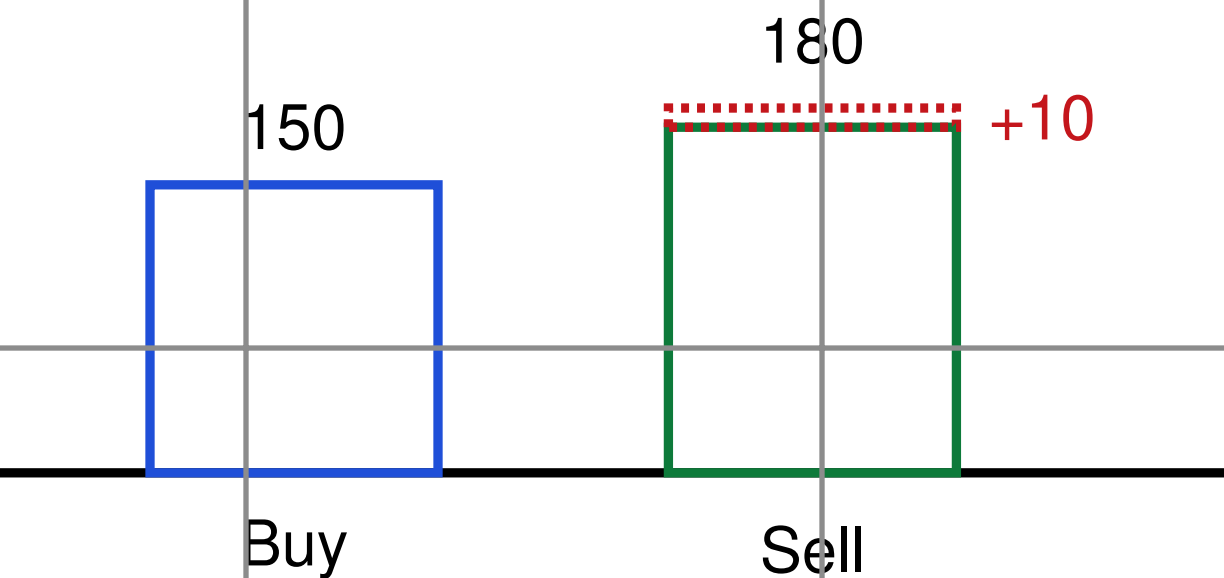
$$\text{Absolute return} = (V_f - V_i) / V_i \times 100$$

$$\text{HPR} = [\text{Income} + (V_f - V_i)] / V_i \times 100$$

$$\text{CAGR} = (V_f / V_i)^{1/n} - 1 \quad (\times 100 \text{ for } \%)$$

Eg: Rs 150 → Rs 180, dividend Rs 10

$$\text{HPR} = (10 + 30) \div 150 = 26.67\%$$



2. Uneven Cash Flows

Unequal cash flows: discount each to today.

$$\text{NPV} = \sum \frac{C_t}{(1+k)^t} - I$$

C_t = cash flow, k = discount rate, I = investment

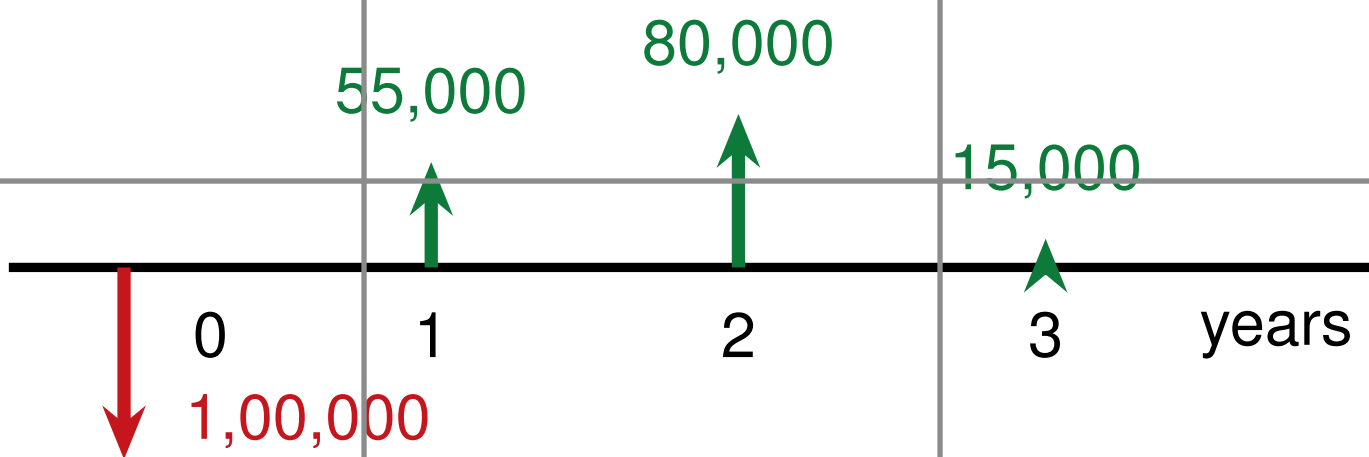
NPV > 0: accept; NPV < 0: reject.

IRR = rate where NPV = 0; accept if IRR > k .

Eg: I = Rs 1,00,000, k = 10%

NPV ≈ Rs 27,340 > 0, so accept

(PVIF 0.909, 0.826, 0.751)



3. Compounding Frequency

More frequent interest = a little more money.

$$A = P \left(1 + \frac{r}{n} \right)^{nt} \quad \text{Continuous: } A = Pe^{rt}$$

$$\text{Effective annual rate} = \left(1 + \frac{r}{n} \right)^n - 1$$

Eg: Rs 5,00,000 at 8% for 1 year

Compounding	Amount (Rs)	EAR
Yearly	5,40,000	8.00%
Quarterly	5,41,216	8.24%
Monthly	5,41,500	8.30%
Daily	5,41,639	8.33%
Continuous	5,41,644	8.33%

4. Economic Equivalence

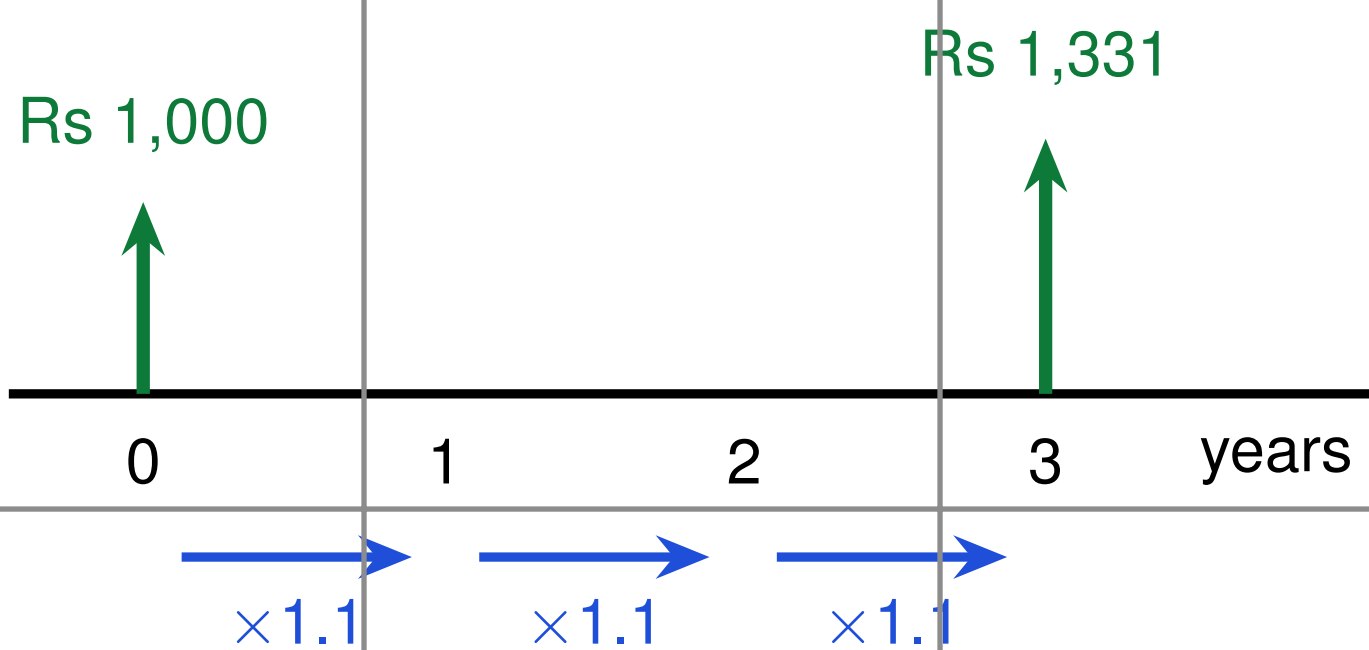
Same value at a common point in time.

$$FV = PV(1+i)^n$$

$$PV = \frac{FV}{(1+i)^n}$$

Eg: at 10%, Rs 1,000 today ≡ Rs 1,331 after 3 years

Use: compare loans, lease vs purchase.



UNIT VI: RISK AND UNCERTAINTY

1. Decision under Risk and Uncertainty

Risk = chance of an undesirable event that causes a financial loss.

Risk management = identify, assess and control risk.

Types of risk:

Strategic

Compliance

Operational

Financial

$$\text{Expected return } \bar{X} = \sum X_i P(X_i)$$

$$\text{Variance } \sigma^2 = \sum d^2 / n \quad (\text{or } \sum d^2 P)$$

$$\text{SD } \sigma = \sqrt{\sigma^2}, \quad d = X_i - \bar{X}$$

$$\text{CV} = \frac{\sigma}{\bar{X}} \times 100$$

Eg: returns 12, 18, 20, 15, 25 (%)

mean 18%, σ = 4.43%, CV = 24.6%

Higher σ or CV = riskier. (σ uses ÷ n)

2. Risk Premium

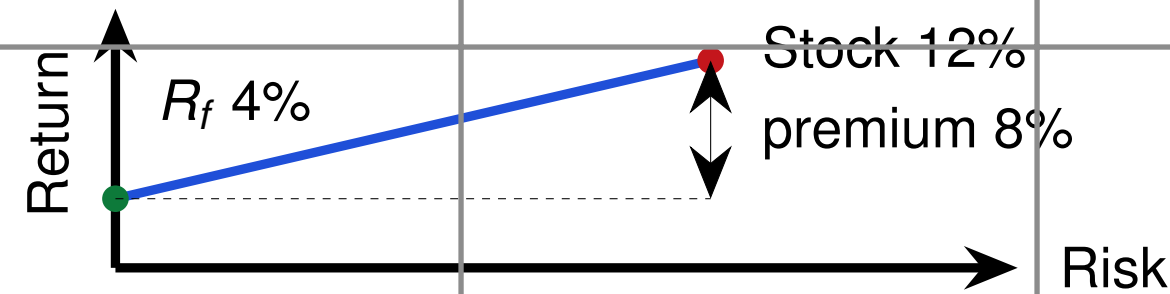
Extra return for taking risk (vs risk-free).

$$\text{Risk premium} = E(R) - R_f$$

$$\text{Sharpe ratio} = \frac{R_p - R_f}{\sigma_p}$$

Eg: $E(R)$ = 12%, R_f = 4%, σ = 10%

Premium = 8%, Sharpe = 0.8



4. Life Insurance

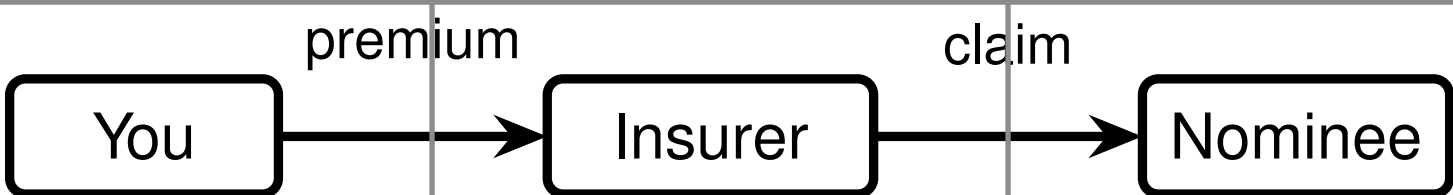
You pay a premium. The insurer pays the sum assured S if you die within the term.

$$\text{APV} = S \sum_{k=1}^n v^k P(T = k), \quad v = \frac{1}{1+i}$$

Eg: 3-yr term, S = Rs 2,00,000, i = 5%

$P(T = 1, 2, 3) = 0.003, 0.0025, 0.002$

APV = Rs 1,370.48



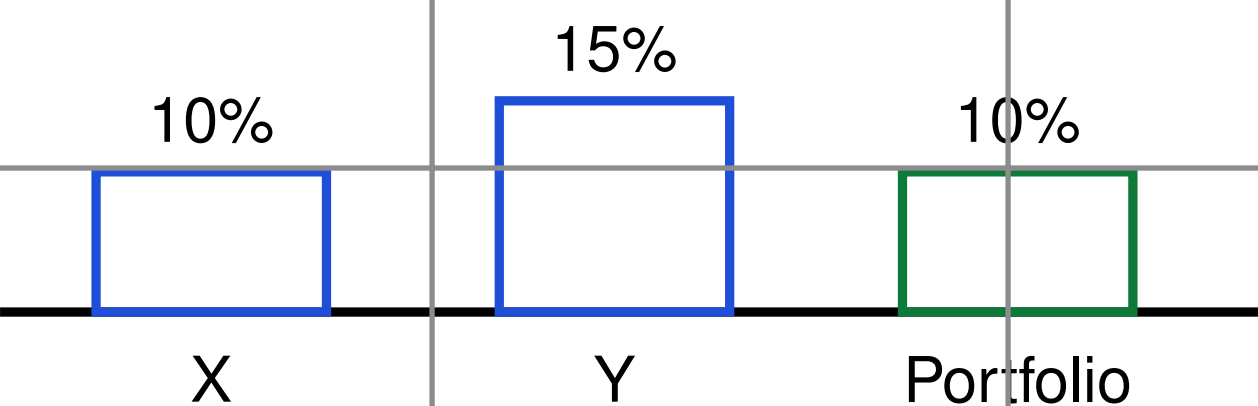
3. Portfolio Diversification

Spread over assets with $\rho < 1$ to lower risk.

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho_{12}$$

Eg: w = 0.5, 0.5; σ = 10%, 15%; ρ = 0.25

σ_p^2 = 0.01, so σ_p = 10%



5. Endowment

Pays the sum assured S if you die in the term OR survive to maturity (protection + savings).

$$\text{APV} = S \left[\sum_{k=1}^n v^k P(T = k) + v^n {}_n P_x \right]$$

Eg: 20-yr plan, S = Rs 1,00,000:

Rs 1,00,000 paid on death or on survival

${}_n P_x$ = chance of surviving n years



Investors demand a higher expected return for taking higher risk